AEROMESH MISSION DECISION REPORT

STATUS: MESH_GENERATED

Phase 5 Drone Validation Mission

Generated: 2026-09-08

SCIENTIFIC ACCURACY & INTEGRITY DISCLOSURE:
- **Coordinate System:** LOCAL_ARBITRARY (Monocular camera optical reference).
- **Scale Status:** RELATIVE_SCALE (No metric ground coordinates without validated calibration).
- **Georeferencing:** UNREFERENCED (WGS84 / GPS anchors not bound; GeoJSON disabled).
- **Reconstruction:** Authoritative sparse SfM; dense MVS was unexecuted due to hardware constraints and no synthetic dense points were fabricated.

1. Mission & Source Video Summary

Mission ID	phase5_drone_validation	**Operator**	AeroMesh Inspection Team
Location / Sector	Operational Flight Zone	**Mission Type**	Infrastructure
Video Source	WhatsApp Video 2026-09-01 at 11.27.02 (1).mp4	**Resolution**	3840x2160
Frame Rate	24.0 FPS	**Duration / Frames**	30.21 s (725 frames)

2. AI Detection & Tracking Performance

Detector Model	yolo11n (yolo11n-official)	**Tracker**	Ultralytics persistent ByteTrack
Total Detections	399	**Unique Tracks**	23
Confidence Stats	Mean: 0.495 | Min: 0.351 | Max: 0.707	**Sampling Rate**	2.0 FPS (61 frames)
Detections by Class	car: 383, truck: 1, train: 15	**Tracks by Class**	car: 21, truck: 1, train: 1

3. 3D Photogrammetry & Surface Reconstruction

SfM Camera Model	SIMPLE_PINHOLE	**Registered Cameras**	20 / 20 (100%)
Sparse Points	12,916	**Mean Reprojection Error**	0.9785 px
Surface Mesh Status	AVAILABLE (pycolmap_poisson)	**Mesh Complexity**	28,139 vertices | 56,120 faces
Dense Reconstruction	UNAVAILABLE (0 synthetic points)	**Coordinate Framework**	LOCAL_ARBITRARY / RELATIVE_SCALE

4. AI-to-3D Multi-View Spatial Fusion

Authoritative 2D Tracks	23	**Mean Reproj Error**	2.393 px (threshold: 25.0 px)
Tracks Evaluated	3	**Acceptance Rate**	100.0%
Association Status	VALID: 1 | LOW_CONF: 1 | INSUFFICIENT: 1 | REJECTED: 0	**Motion States**	STATIC: 3 | MOVING: 0

Object ID	Track	Class	Motion	Status	Position (X, Y, Z)	Reproj (px)
OBJ_T0001	T0001	car	STATIC	VALID	[-17.52, -5.48, 145.64]	1.95
OBJ_T0002	T0002	car	STATIC	INSUFFICIENT_EVIDENCE	[-13.27, -7.65, 152.11]	N/A
OBJ_T0020	T0020	car	STATIC	LOW_CONFIDENCE	[-18.48, 0.25, 148.01]	8.02

5. Geometric Measurements & Metric Scale Calibration

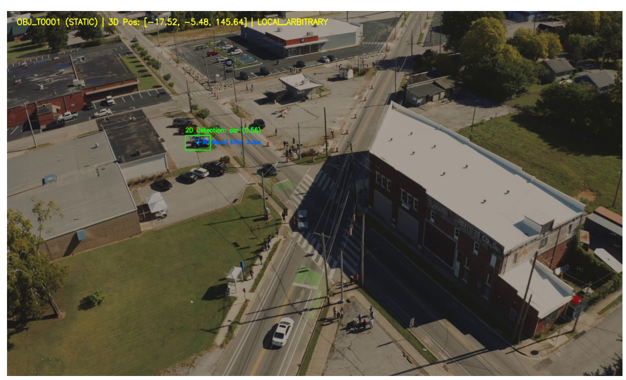
Calibration ID	CAL_phase5_drone_validation_1788531696	**Method**	KNOWN_REFERENCE_DISTANCE
Scale Factor	2.39036 m/unit	**Known Baseline**	15.0 m (Confidence: 0.95)

Label / Type	Value	Unit	Status	Confidence	Uncertainty
Vehicle Baseline Distance (Calibrated)	15.00	m	METRIC_CALIBRATED	0.95	±0.15
Vehicle Baseline Distance (Relative Uncalibrated)	6.28	relative_units	RELATIVE	0.90	N/A
Vehicle Dimensions OBJ_T0001 (Calibrated)	L: 4.54, W: 2.15, H: 1.67	m	METRIC_CALIBRATED	0.85	±0.12
Surface Mesh Volume Check	Volume computation refused for open surface mesh (not watertight).	m³	REFUSED_NON_WATERTIGHT	0.00	N/A

6. Visual Evidence & Reprojection Overlays

The following imagery represents authoritative multi-view observations reprojected onto the 3D model with 2D bounding boxes:

Overlay: overlay_OBJ_T0001_frame_00000.jpg



Overlay: overlay_OBJ_T0001_frame_00001.jpg

7. Comprehensive Mission Limitations & Disclosures

• LOCAL_ARBITRARY: Reconstruction coordinates are arbitrary relative units, not true meters or GPS coordinates.
• RELATIVE_SCALE: Monocular video SfM is scale-ambiguous; metric distances are strictly valid only where verified ground scale calibration is applied.
• UNREFERENCED: Scene is unreferenced against EPSG/WGS84. GeoJSON geographic coordinates (latitude/longitude) are unavailable.
• DENSE_MVS_UNAVAILABLE: Dense stereo reconstruction requires CUDA or HIP hardware. Sparse SfM is preserved as authoritative geometry without synthetic point fabrication.
• UNOBSERVED_SURFACES: Unobserved, occluded, or low-parallax areas are preserved as unobserved rather than interpolated as synthetic geometry.